**3.Write a Pandas program to display the details of jobs in descending sequence on job**

**title.**

**Aim**

To write a Python Pandas program to read a jobs dataset and display the details of jobs in **descending order based on Job Title**.

**Algorithm**

1. Import the Pandas library.
2. Load the jobs dataset into a Pandas DataFrame.
3. Display the original job details.
4. Sort the DataFrame using the **Job Title** column in descending order.
5. Display the sorted job details.
6. End the program.

**Dataset**

| Job_ID | Job Title | Company | Location | Salary |
|---|---|---|---|---|
| J001 | Data Analyst | ABC Ltd | Chennai | 45000 |
| J002 | Software Engineer | XYZ Tech | Bangalore | 60000 |
| J003 | Data Scientist | DataCorp | Chennai | 75000 |
| J004 | Business Analyst | Infosys | Hyderabad | 55000 |
| J005 | Machine Learning Engineer | TCS | Pune | 70000 |
| J006 | Web Developer | Wipro | Chennai | 50000 |
| J007 | AI Engineer | Google | Bangalore | 90000 |

**Code:**

C: > Users > DELL > Desktop > dsa0404 4_files > set-2(3).py > ...

```python
import pandas as pd
# Read the dataset
jobs = pd.read_excel(r"C:\Users\DELL\Desktop\dsa0404 4_files\job.xlsx")
# Display original data
print("Original Job Details:")
print(jobs)
# Sort by Job Title in descending order
sorted_jobs = jobs.sort_values(by="Job Title", ascending=False)
# Display sorted data
print("\nJob Details in Descending Order of Job Title:")
print(sorted_jobs)
```

```
  Job_ID                  Job Title   Company   Location  Salary
5   J006              Web Developer     Wipro    Chennai   50000
1   J002          Software Engineer  XYZ Tech  Bangalore   60000
4   J005  Machine Learning Engineer       TCS       Pune   70000
7   J008     Database Administrator       IBM     Mumbai   65000
2   J003             Data Scientist  DataCorp    Chennai   75000
0   J001               Data Analyst   ABC Ltd    Chennai   45000
3   J004           Business Analyst   Infosys  Hyderabad   55000
6   J007                AI Engineer    Google  Bangalore   90000
```

## Result

Thus, the Pandas program was successfully executed and the job details were displayed in descending order of Job Title.

## 4. Pandas Program to Create a Line Plot of Historical Stock Prices of Alphabet Inc.

## Aim

To write a Python Pandas program to read the historical stock price data of **Alphabet Inc. (GOOGL)** and create a **line plot** of its closing stock prices between two specific dates.

## Algorithm

1. Import Pandas and Matplotlib.
2. Load the historical Alphabet stock dataset.
3. Convert the Date column into Pandas datetime format.

4. Specify the starting and ending dates.
5. Filter the stock data between the specified dates.
6. Plot the Date against the Closing Price.
7. Add title, axis labels, grid and legend.
8. Display the line plot.
9. End the program.

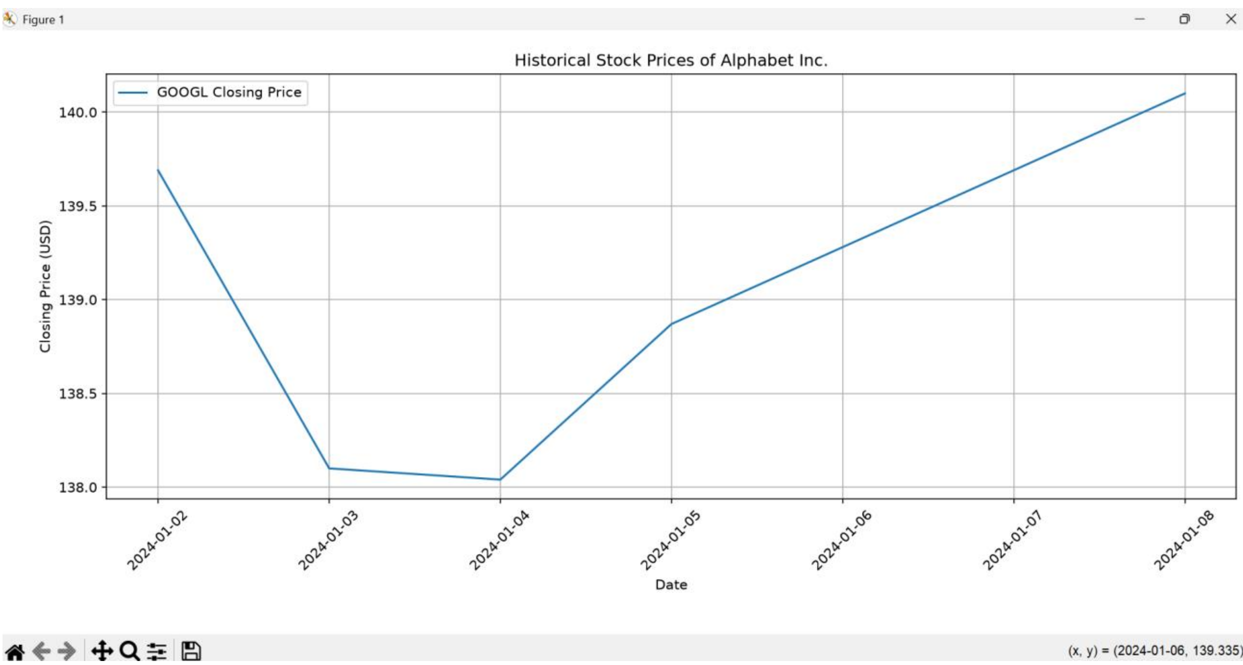
**Dataset**

| Date | Open | High | Low | Close | Volume |
|---|---|---|---|---|---|
| 2024-01-02 | 138.55 | 140.62 | 138.41 | 139.69 | 20000000 |
| 2024-01-03 | 138.60 | 139.94 | 137.74 | 138.10 | 18000000 |
| 2024-01-04 | 138.12 | 139.50 | 137.78 | 138.04 | 17000000 |
| 2024-01-05 | 138.35 | 139.90 | 138.20 | 138.87 | 16000000 |
| 2024-01-08 | 139.00 | 141.20 | 138.80 | 140.10 | 19000000 |

```python
import pandas as pd
import matplotlib.pyplot as plt
# Read the Alphabet stock dataset
stock = pd.read_excel(r"C:\Users\DELL\Desktop\dsa0404 4_files\alphabet.xlsx")
# Convert Date column to datetime
stock["Date"] = pd.to_datetime(stock["Date"])
# Specify the date range
start_date = "2024-01-01"
end_date = "2024-12-31"
# Filter data between the two dates
filtered_stock = stock[
    (stock["Date"] >= start_date) &
    (stock["Date"] <= end_date)
]
# Create line plot
plt.figure(figsize=(10, 5))
plt.plot(
    filtered_stock["Date"],
```

```
    (stock["Date"] >= start_date) &
    (stock["Date"] <= end_date)
]
# Create line plot
plt.figure(figsize=(10, 5))
plt.plot(
    filtered_stock["Date"],
    filtered_stock["Close"],
    label="GOOGL Closing Price"
)
plt.title("Historical Stock Prices of Alphabet Inc.")
plt.xlabel("Date")
plt.ylabel("Closing Price (USD)")
plt.grid(True)
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

## Output:




## Result

Thus, the Pandas program was successfully executed to filter Alphabet Inc. historical stock prices between two specified dates and create a line plot of its closing prices.